

SIMATS ENGINEERING

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Course: MLA0401
DEEP LEARNING

Question 1: Medical Image Classification using CNN

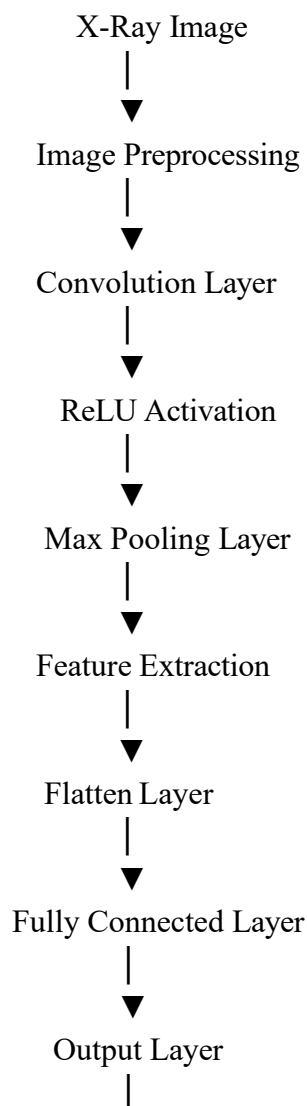
Question

Explain how a Convolutional Neural Network (CNN) is used for Medical Image Classification. Describe the role of convolution, pooling, fully connected, and output layers.

A)Explanation

Medical Image Classification is a Deep Learning application that automatically identifies whether an X-ray image is Normal or Abnormal. CNN extracts important features such as edges, bones, and disease patterns using convolution layers. Pooling layers reduce the image size while preserving important information. The fully connected layer analyzes these features, and the output layer predicts the correct class.

Workflow





Normal / Abnormal Prediction

Question 2: Face Recognition using CNN

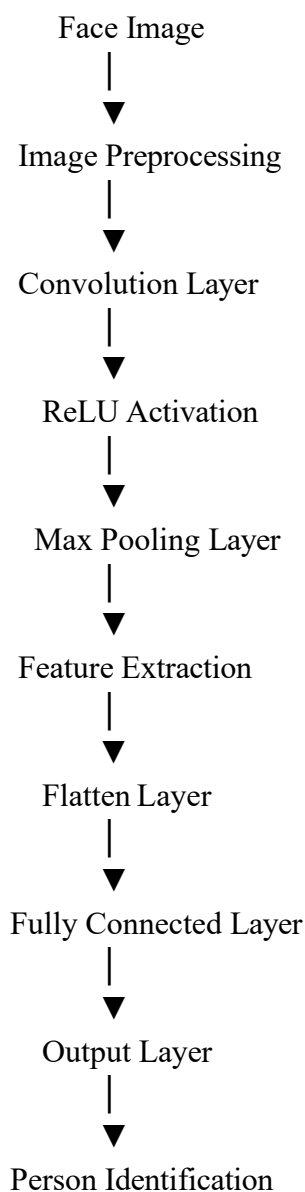
Question

Explain how a Convolutional Neural Network (CNN) is used for Face Recognition. Describe the role of convolution, pooling, fully connected, and output layers.

A)Explanation

Face Recognition is a Deep Learning application that identifies people using facial images. CNN learns facial features such as eyes, nose, mouth, and face shape. Convolution layers extract facial features, pooling layers reduce image size, and the fully connected layer analyzes the extracted information. The output layer predicts the identity of the person.

Workflow



Question 3: Autonomous Vehicle Image Recognition using CNN

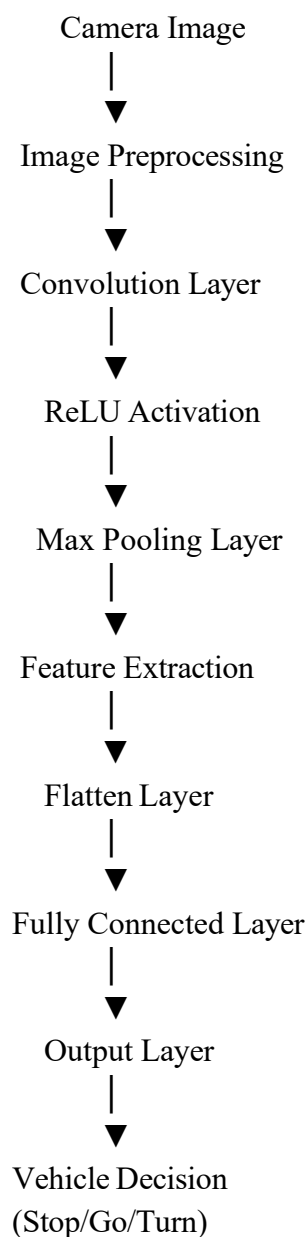
Question

Explain how a Convolutional Neural Network (CNN) is used in Autonomous Vehicles for Image Recognition. Describe the image recognition process and the role of different CNN layers.

A) Explanation

Autonomous vehicles use CNN to recognize road signs, traffic lights, pedestrians, and other vehicles. The convolution layers extract important road features, pooling layers reduce image size, and the fully connected layer analyzes the detected objects. The output layer identifies the object, helping the vehicle make safe driving decisions.

Workflow



Question 4: Overfitting in CNN

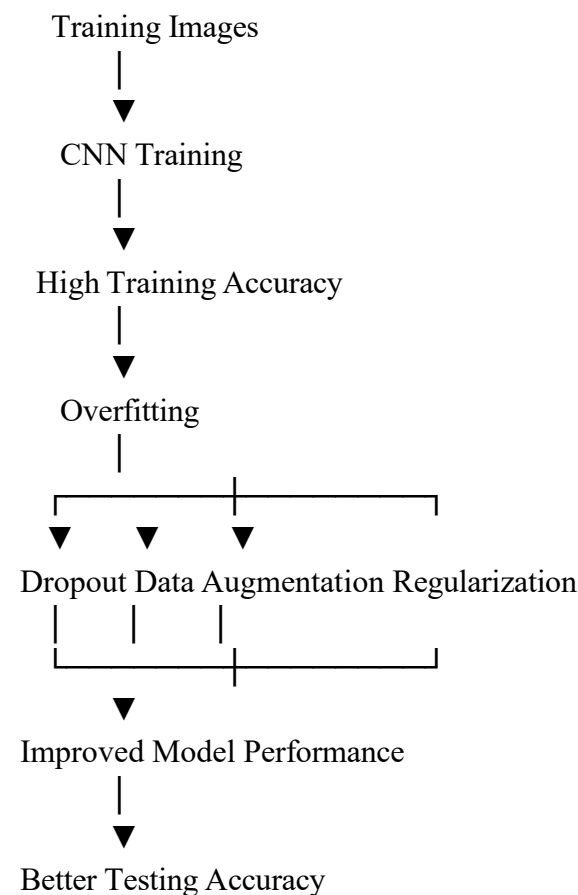
Question

Explain Overfitting in Convolutional Neural Networks (CNN). Describe how Dropout, Data Augmentation, and Regularization improve CNN performance.

A)Explanation

Overfitting occurs when a CNN learns the training data too well but performs poorly on new data. The model memorizes the training images instead of learning general patterns. Techniques such as Dropout, Data Augmentation, and Regularization help reduce overfitting and improve the model's testing accuracy.

Workflow



Question 5: Fruit Image Classification using CNN

Question

Explain how a Convolutional Neural Network (CNN) is used for Fruit Image Classification. Describe the image classification process and the role of different CNN layers.

A)Explanation

Fruit Image Classification is a Deep Learning application where CNN identifies fruits such as Apple, Banana, Orange, and Mango. The convolution layers extract features like color, shape, and texture. Pooling layers reduce image size while preserving important information. The fully connected layer classifies the extracted features, and the output layer predicts the correct fruit category.

Workflow

